

MACHINE LEARNING-BASED CELL-FREE SUPPORT IN THE O-RAN ARCHITECTURE: AN INNOVATIVE CONVERGED OPTICAL-WIRELESS SOLUTION TOWARD 6G NETWORKS

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ABSTRACT

One of the most crucial challenges for the global socio-technical domain nowadays is the support of unprecedented use cases for a variety of applications such as augmented reality, 3D holographic telepresence, Industry 4.0, robotics, e-health, and pervasive connectivity. In this environment, beyond 5G intelligent networks that are able to provide enhanced flexibility through the dynamic allocation of the system's resources, while realizing the perceived zero-latency vision, are envisioned to ensure a smooth transition toward 6G. The proposed innovative converged optical-wireless network configuration targets to enable such a transformation, through the development of novel radio access networking solutions. These advanced radio-edge architectures are designed by utilizing the distributed cell-free concept and the serial fronthaul approach, while contributing innovative functionalities to the O-RAN project. At the same time, the proposed approach facilitates integrated connectivity of both mobile and fixed services, which share the same edge and midhaul infrastructures, while both are served by a core. The proposed network architecture can be envisioned as the key technology enabler to satisfy the requirements of future 6G networks through the modifications of the O-RAN interfaces toward a cell-free-enabled solution, and through the incorporation of novel Machine Learning techniques at the Radio Edge.

INTRODUCTION

Communication networks currently undergo a number of structural and functional changes that are associated with the need to support current and future demands (e.g., mobile traffic is expected to grow three-fold within 2025, as a result of predicted 10 million devices per km² and 100 billion devices worldwide), and with the commitment for the provision of ultra-low latency and reliability [1]. With the advancement of technology, the communication nodes will be inseparable from human social activities. To truly realize the connection and interaction needs at any-time and anywhere, future networks need to have specific characteristics, such as:

- Omnipresence (covering air, ground, space and sea)
- All-in-one (Internet of Everything)
- Omniscience (with the help of various sensors)
- All-purpose (based on big data and deep learning)

The same circumstances are also forecasted for other fields, such as Industry 4.0, health sectors, manufacturing industries, and autonomous industries, where revolutionary changes are underway. Another major challenge of the future 6G networks is to reach climate goals by reducing CO₂ emissions. This will be achieved through improved operational and management efficiency and energy consumption reduction [2].

To tackle these challenges, several steps should be taken from the perspective of network concepts and technological aspects, in order to ensure economic developments, reliability, and increased energy efficiency. Key advances are required both at the network design and network management levels to accommodate highly dynamic traffic demand, achieve energy reduction, and provide cost-effective multi-level resource pooling. Specifically, the network should support multiple distributed edge nodes and a multitude of access points coordinated by entities in a low-cost and near-zero latency manner. A unified and hierarchical infrastructure is essential to provide intelligent management of communication, computation and storage resources. The incorporation of efficient Machine-Learning (ML) algorithms can further improve network efficiency. Additionally, infrastructure sharing is a sustainable way of lowering the network's capital and operating expenditures. Support of multiple tenants is the key enabler of the new sharing economy business models in the telecommunication industry. Novel mechanisms for resourceful and efficient management of shared networks are necessary in order to enable a plethora of use-cases and industry verticals targeted in beyond-5G and 6G systems [3].

Following the aforementioned challenges, in this article, we present an innovative network configuration that integrates wireless, optical, and computation resources, targeting flexible connectivity of a massive number of antennas, and aiming to unlock

the potential of user-centric Cell-Free (CF) deployments in 6G networks. Our solution aims to overcome the isolation and underutilization of resources deployed at edge nodes, and offers zero perceived latency to innovative connectivity applications. The main idea behind the proposed network architecture design is the utilization of a converged optical-wireless X-haul infrastructure that considers scalable CF networks at the radio edge. Specifically, Radio Units (RUs) are interconnected with CF Distribution Units (DUs) via fronthaul links based on a bus configuration (i.e., Radio Stripes, [4]), while the support of 10-Gigabit-Symmetrical (XGS)- Passive Optical Network (PON) backhauling ensures the reliable support of dense RU deployment. In addition, the proposed approach is fully aligned with the Fixed-Mobile Convergence (FMC) paradigm, allowing the midhaul optical fibers to be flexibly shared with fixed services. The incorporation of CF allows a large number of RUs to be interconnected in a bus topology with a single DU. This solution increases channel capacity significantly and achieves unprecedented spectral efficiency and performance, which are not constrained by inter-cell interference, thus addressing ultra-dense traffic requirements in an efficient way [5].

Furthermore, our approach is based on the disaggregation of the traditional CF Central Processing Unit (CPU) in DUs and a Central Unit (CU), in line with the 3GPP Next-Generation Radio Access Network (NG-RAN) architecture. This disaggregation of the RAN components defines the splitting of the Next-Generation-NodeB (gNB) protocol stack into Centralized Unit Control Plane (CU-CP), CU-User Plane (CU-UP), and DU nodes connected by corresponding interfaces. The O-RAN specification introduces the Near-Real-Time RAN Intelligent Controller (Near-RT RIC), which implements radio resource management, measurement and telemetry. However, the current O-RAN specification does not provide any CF-supported use-cases to evaluate the RAN performance. The proposed solution extends the O-RAN disaggregation of the BaseBand Unit (BBU) in a virtualized O-RAN Distribution Unit (vO_DU) with the real-time functions, and a virtualized O-RAN Central Unit (vO_CU)) with the non-real time functions, where the CF high-PHY functions (e.g., modulation and precoding) are integrated with the vO_DU, thus adding innovative CF support in O-RAN's architecture. In addition, as the proposed radio-edge architecture is based on multiple RUs that jointly provide connectivity to end-users, RU clustering (and re-clustering) approaches, as well as dynamic adaptability techniques are essential in order to improve the achievable spectral efficiency.

This article is organized as follows: The following section discusses the CF concept, together with challenges that arise from the implementation at the radio-edge. We then present the proposed CF-based fronthaul configuration, and a Q-learning algorithm for the evaluation of the proposed network configuration. Finally, we conclude in the final section.

THE CELL-FREE NETWORKING APPROACH

In the CF networking paradigm, all the single-antenna UEs are served by a vast number of distributed Access Points (APs) connected to a CPU and managed by a centralized processing pool in the backhaul network [6]. Consequently, CF networks eliminate the cell-edge user problem and

provide enhanced connectivity to all users, while significantly improving both energy efficiency and quality of service [7, 8]. Thus, this approach is a highly promising solution for 6G networks due to the unique combination of advantages related to favorable propagation, high macro-diversity, and augmented expected coverage.

The initial CF concept assumed that UEs (sharing the same time-frequency resources) are jointly served by all APs. However, this approach is not scalable, power-inefficient and impractical, since only a fraction of the APs can beneficially transmit to a specific UE. Logically, the CF concept evolved to a user-centric system with UEs being served dynamically by a sub-set (cluster) of APs selected based on the serving distance, channel quality or network performance. However, the optimal AP selection procedure for cluster creation is an open research question. The two key issues that should be addressed are:

- Coordination between APs (directly influencing the fronthaul requirements and distribution of computational power between APs and CPU)
- Inter-CPU coordination if the serving cluster consists of APs connected to different CPUs.

The amount of information shared between APs and a CPU is a critical consideration for CF networking. In fact, the achieved Spectral Efficiency (SE) is a function of the level of coordination (i.e., the amount of information shared) between APs. For example, it is almost doubled when Channel State Information (CSI) is sent to the CPU for performing joint combining/precoding. The study in [9] showed that the highest SE is achievable in a fully centralized network, where the pilot and data signals are transmitted to the CPU via fronthaul links, while the CPU performs the channel estimation and data detection procedures. Unfortunately, such a solution is not scalable since the fronthaul requirements grow with the number of APs.

Radio Stripes [4] and Radio Weaves [10] were proposed to cope with the scalability issue. In these solutions, more advanced APs (connected in bus and tree topologies, respectively) can combine and forward local channel estimates, CSI and error statistics. This information is used at the next AP for improving the local estimate; the final AP computes the final signal estimate and sends it to the CPU. This sequential processing results in a significant improvement of the system's SE compared to the situation where the APs use only own local estimates [4]. Additionally, it was suggested to tackle the scalability issue by adapting the dynamic cooperation clustering to CF networks [5]. Unfortunately, this work was focused on the case where all APs serving a user belong to the same CPU. It was suggested to avoid inter-CPU coordination by considering disjoint clusters, or treat the combination of all CPUs as one large CPU. The latter approach does not consider the limitations of midhaul links and other practical constraints.

Clusters of APs assigned to different CPUs is a novel paradigm for CF networks; such an approach, however, poses a new set of challenges for coordination and fronthauling. Since CF networks require a high level of adaptability of the resource distribution due to highly dynamic user demand, we suggest that the new CF paradigm must be fully aligned with recent initiatives toward an Open RAN, offering a sufficiently high level of flexibility.

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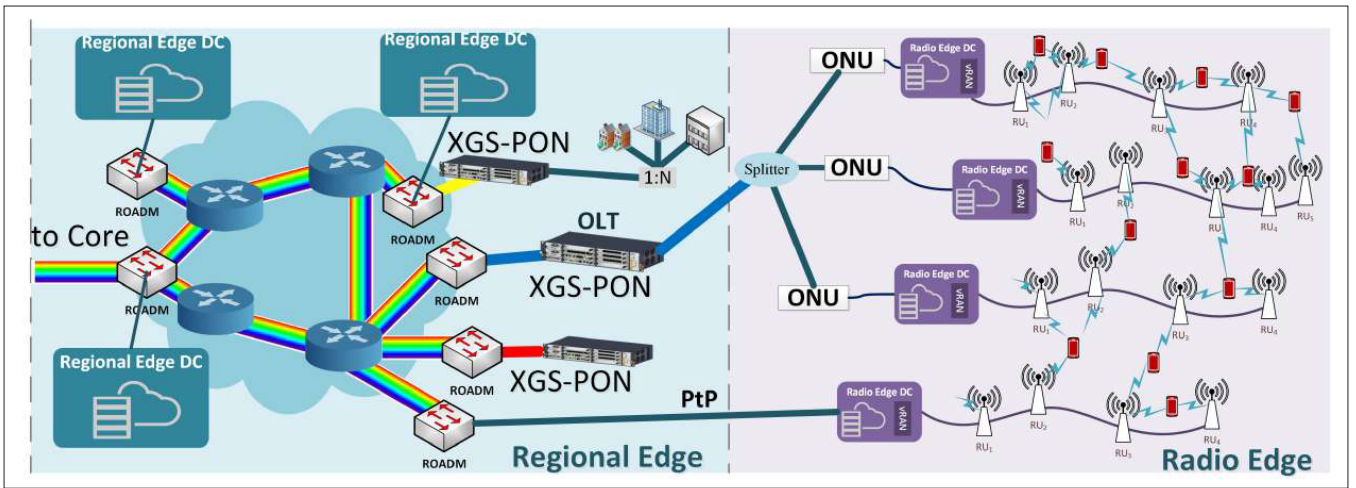


FIGURE 1. The proposed converged optical-wireless network with CF support.

DESIGN OF CELL-FREE BASED FRONTHAUL IN-LINE WITH THE O-RAN ALLIANCE

In the last few years, many fronthauling technologies have been proposed, advocating for the functional split of Next Generation NodeB (gNBs). The fronthaul design for Centralized-RAN (C-RAN) architectures has been associated with two main components, that is, the Remote Radio Heads (RRHs) and the BBUs that are centralized in a remote location. 3GPP has defined eight functional split options, inducing trade-offs between centralization gains and fronthaul bandwidth requirements. In this direction, both digital and analog or Radio over Fiber (RoF) fronthauling solutions have been proposed and extensively studied. In a converged fiber-wireless fronthaul, analog BBUs are fiber-connected to several analog RRHs that leverage massive MIMO to create point-to-multipoint mmWave links [11]. Digital RoF fronthaul also achieves point-to-multipoint connectivity with the additional benefits of offering a packet-based solution with high flexibility and better compatibility with multiple services. However, the C-RAN architecture has fallen short both in required bandwidth and flexibility to meet the ambitious 5G traffic demands. In this context, 3GPP has promoted a new design that enables various functional splits, where BBUs are further disaggregated and virtualized in CUs and DUs that are connected to the RUs. The integration of both fronthaul and backhaul segments into a common transport infrastructure is explored by various initiatives, such as O-RAN that aims to provide an open fronthaul interface, thus enabling multi-vendor DU-RU interoperability. Among the functional split options that can support a packet-based fronthaul, the option 7-2 is a variation of intra-PHY split that has been extensively studied and seems to be a prominent choice [12].

In this light, we propose a novel converged optical-wireless configuration based on the CF networking solution, that targets the flexible connectivity of a massive number of RUs, aiming to promote user-centric CF deployments in 6G networks. The proposed approach is illustrated in Fig. 1 and consists of an evolved RAN that is extended with emerging CF technologies, as well as of an innovative optical transport domain. The latter deploys a distributed edge infrastructure with Data Centers

(DCs) structured in 2 tiers, featuring Regional Edge and Radio Edge nodes. In the following subsections, we analyze the proposed networking solution by detailing the innovative research challenges in the RAN and fronthaul/midhaul domains, as well as in the regional edge, while also discussing the requirements for extending the O-RAN architecture in order to support CF networking.

CF RAN WITH DATA-DRIVEN, FULLY DISTRIBUTED PROCESSING

At the radio edge, the proposed radio network configuration aims at addressing the most pressing CF limitations that will allow CF deployment in future 6G RANs, based on the utilization of dynamic cluster formation algorithms. The dynamic feature of such RU clustering algorithms is based on the fragmentation of the CF CPU in multiple DUs. This procedure triggers the support of distributed computation and coordination between RUs and between DUs. Under this networking configuration, clusters of RUs connected to multiple DUs jointly address inter-DU and RU-DU coordination for the first time, while also considering the constraints introduced by fronthaul and midhaul links. The optimal cooperation levels between RUs and DUs and between DUs can be guaranteed through the application of dynamic adaptability algorithms of the involved entities' coordination levels.

Going a step further, the application of data-driven dynamic cluster formation algorithms are able to provide more efficient results when the training procedure is based on real channel data. Such an approach is considered in our network by leveraging ML-based clustering techniques using real channel measurements, targeting real-time operation leveraging peak data rate as the loss function. Firstly, by definition, the measured channel data is free of unrealistic channel modeling assumptions. Secondly, the constraints imposed by the realistic CF networks require joint solving of a many optimisation problems. Note that each optimisation problem is solved in isolation (i.e., not taking into account constraints imposed by other problems). This divide and conquer approach is analytically tractable, but it can lead to suboptimal results when a complete system is considered.

In parallel, for the development of RU clustering algorithms we consider mobile (i.e., non-static) users, so, at some point, the cluster must be

A complementary distributed RRM solution can be also applied as a part of the vO_DU, which can locally schedule resources of the underlying infrastructure at the radio edge node, and complement the global view provided at the level of Near RT-RIC.

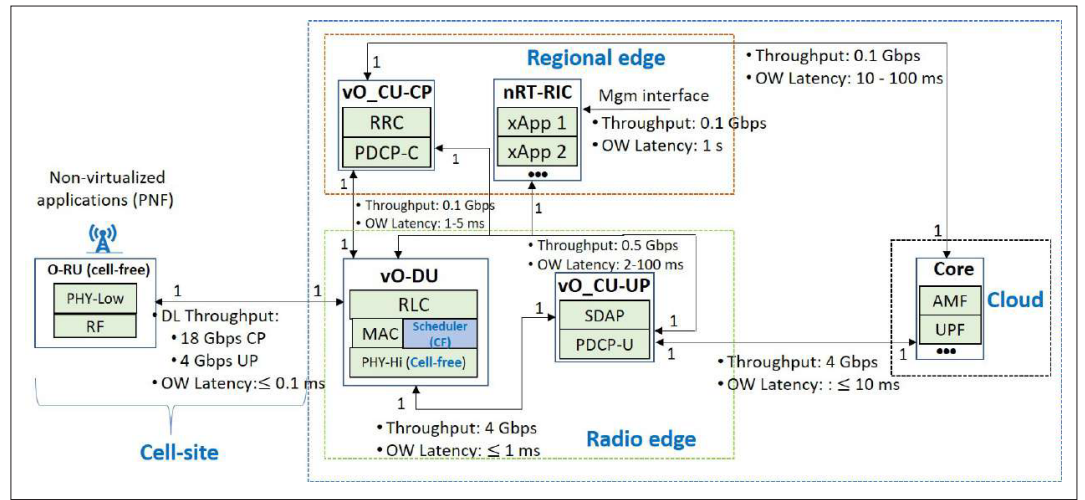


FIGURE 3. Option 7.2x for the proposed cell-free-based network architecture.

architecture with multiple CF-based innovations. Precisely, at the O_DU module a distributed CF is incorporated, by properly modifying the PHY, MAC, and RLC sublayers. The role of orchestration in the midhaul (Network Function Virtualization Orchestrator (NFVO), Mobile Edge Computing (MEC)) is also considered, as discussed in the next paragraph. Finally, the CF Radio Resource Management (RRM) is also considered in the proposed architecture as part of the Near-RT RIC, in line with the O-RAN specifications for training and deploying ML inference within the Near-RT RIC. A complementary distributed RRM solution can be also applied as a part of the vO_DU, which can locally schedule resources of the underlying infrastructure at the radio edge node, and complement the global view provided at the level of Near RT-RIC. Finally, a Near-RT Software Defined Networking (SDN) function is also required that can be hosted by the Near-RT RIC, disaggregating the Non-RT SDN control and policy function currently offered by the O-RAN stack. This configuration targets to support near real-time reaction to workload variations, at sub-second timescales.

OPTICAL NETWORK FOR MIDHAUL TRANSPORTATION AND FIXED-MOBILE INTEGRATION

The proposed converged optical-wireless network considers an optical midhaul network, which interconnects regional edge nodes through a ring configuration, via Reconfigurable Optical Add Drop Multiplexers (ROADMs), as depicted in Fig. 1. Each node incorporates XGS-PON and PtP interfaces, offering respectively, a shared fiber with Time Division Multiple Access (TDMA) and either direct fiber or a shared fiber with Wavelength Division Multiplexing (WDM) optics. This configuration also delivers a solution for FMC, allowing access capacity to be shared by fixed applications (e.g., Fiber To The Home (FTTH)) and mobile applications (e.g., transportation of radio edge traffic from the E2 and E1 interfaces). Both fixed and mobile users share the same midhaul and edge infrastructures (e.g., MEC hosts) and are both served by the core. FMC is implemented via Point-to-Point (PtP) Ethernet and Point-to-Multi-Point (PtMP) smart XGS-PON Small-Form-factor-Pluggable (SFP) – Optical Line Terminal (OLT) interfaces deployed at the same regional edge node, allowing capacity to be shared by PtMP links or PtP links, as illustrated in Fig. 4.

This optical access architecture offers a high degree of flexibility, allowing capacity to be dynamically shared by fixed and mobile clients, leveraging on the aforementioned hierarchical control plane. Specifically, the Software Defined Transport Network (SDTN) controller can flexibly control traffic aggregation at the Regional Edge node and dynamically configure various parameters at the SFP-OLT level (e.g., related to timing and queue management) via RESTCONF messages. Thus, during light loads of the PtMP segment, its unused capacity can be diverted to the PtP segment, while the SDTN controller triggers the dynamic reconfiguration of the RAN sub-slice to exploit this capacity, directly interacting with the Near-RT SDN function via an SDNi interface.

At the regional node, traffic aggregation is implemented via Open vSwitch (OVS), under the control of an edge-deployed SDTN controller. The SDTNs at the edge are coordinated by the Network-Slicing-as-a-Service (NSaaS) subsystem at the core-tier NFVO, and communication is established for this task via the OR-Wi interfaces. Based on this configuration, intelligent traffic management and load balancing among PtP and PtMP traffic via SDN can be implemented, while improved power consumption can be achieved via traffic re-routing and shutting down individual SFP-OLTs during light loads. Additionally, the implementation of predictive slice reconfiguration that provide traffic fluctuations' prediction can trigger slice reconfiguration, thus increasing statistical multiplexing while minimizing the impact on Service Level Agreements (SLAs).

Q-LEARNING-BASED EVALUATION OF THE PROPOSED NETWORK ARCHITECTURE

In this section, we evaluate the effectiveness of the proposed network architecture, in terms of the downlink spectral efficiency. To this end, our simulation environment consists of 64 RUs with antennas mounted 10m above the ground level, and of 10 UEs. The RUs are distributed in two rows simulating lamp-posts along a street. The carrier's frequency is 4 GHz with 10 Physical Resource Blocks (PRBs), while the simulator's channel model is based on the ITU-R M.2412-0 Report (slow fading) and the ITU-R M.2135-1-2009 Report (fast fading). The Round Robin model is our reference model, where each UE is served by its strongest AP. Two different pre-

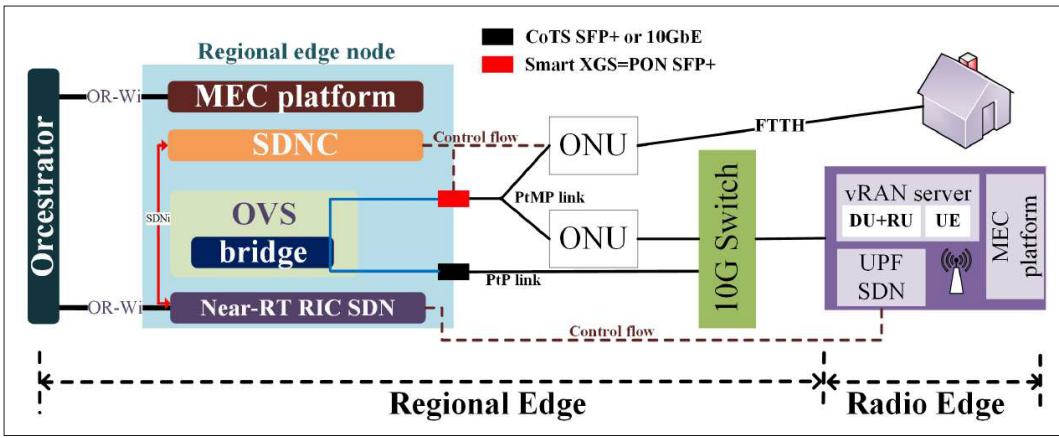


FIGURE 4. Converged optical-wireless configuration and distributed control plane.

coders are used in the simulator for the CF mMIMO networks at the radio edge, that is, Maximum Ratio Transmission (MRT) and Zero Forcing (ZF). For both precoders, the Q-learning algorithm is used to find an optimum number of RUs serving 10 UEs.

The Q-learning algorithm starts from a randomly selected set of RUs and performs iterative RUs' reallocations to find an optimized set of serving RUs. The RUs' reallocations are based on the Q-function, which stores the knowledge about the system obtained during the learning process. The latter process has been performed for MRT and ZF precoders, and for the given channels between the UE and RUs. To limit the algorithm complexity, we assume PRB independence, and we determine the set of serving APs separately for each of the PRBs in the carrier bandwidth. Since the channels on each PRB are different, the training process has been performed separately for each PRB. The performance of the algorithm strongly depends on the definition of the three Q-learning associated concepts: a state, an action and a reward. In our approach, the state is a set of serving RUs, the action is the RU reallocation, while the reward is the increase of the system spectral efficiency resulted from the action. To further limit the algorithm complexity, the state space has been reduced by not storing information of which RUs perform well together, but rather which RUs perform well with any other RUs. This approach results in fast learning, but does not exploit all possible sets of serving RUs.

It should be noted that for the practical deployment of the proposed Q-learning, the trained model of the algorithm can be located at the regional edge and the actual decision model can be located at the radio edge data center. To reflect the performance of the proposed algorithm, we evaluate the performance of per user spectral efficiency and downlink system capacity. The coverage, defined as the spectral efficiency provided for 95 percent of UE can be derived from Fig. 5, where it corresponds with a 0.05 Cumulative Density Function. The Round Robin provides coverage of 1.38 b/s/Hz, the MRT provides 0.32 b/s/Hz while the ZF offers 5.93 Mb/s/Hz. Even though the MRT provides better average UE throughput compared to the SISO, the provided coverage is weaker. The CF ZF with RL shows better performance compared to the CF MRT with RL and to the Single-Input-Single-Output (SISO) round robin algorithms.

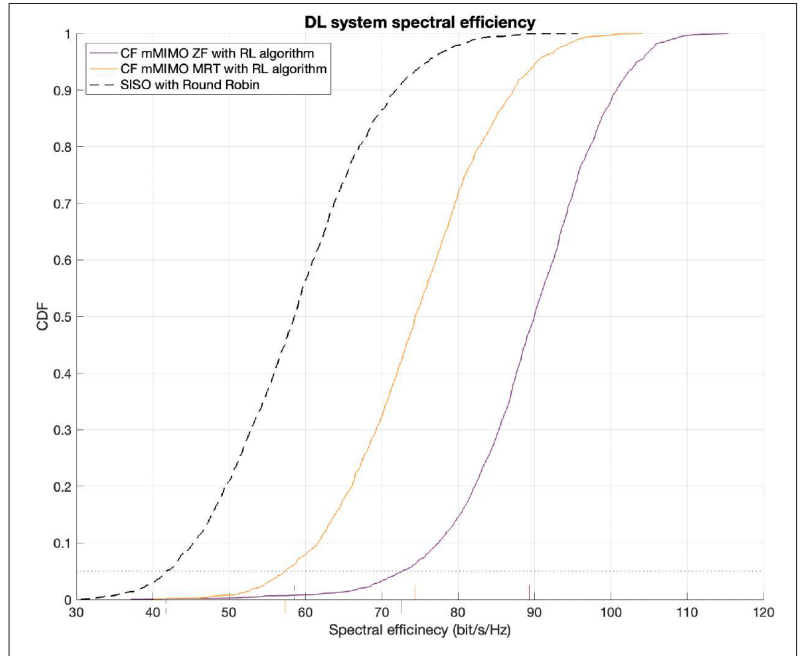


FIGURE 5. Per user spectral efficiency performance of the proposed Q-learning algorithm.

From the network provider's point of view, the network capacity or the system's SE is an important aspect. It can be seen in Fig. 6 that CF MRT provides much higher system SE of 74.3 Mb/s/Hz compared to the SISO Round Robin (58.6 Mb/s/Hz). At the same time, when the CF MRT provides higher system spectral efficiency compared to the SISO Round Robin, the coverage of the CF MRT is weaker. This is caused by the CF MRT offering high SE to UEs located in good radio conditions at the coast of lower SE of UEs located in poorer radio conditions.

The CF ZF is far better than the SISO Round Robin and CF MRT offering 89.4 Mb/s/Hz system SE, even though the simulating average external interference level was relatively high and equal to -90 dBm per subcarrier. Thus, the ZF appears to be an attractive precoding approach in CF mMIMO. Also, other precoders like Regularized Zero Forcing (RZF) need to be investigated together with their impact on the interference level. The RZF precoder, compared to the ZF, improves the received signal power by transmitting more power which also results in more interference therefore advantages of

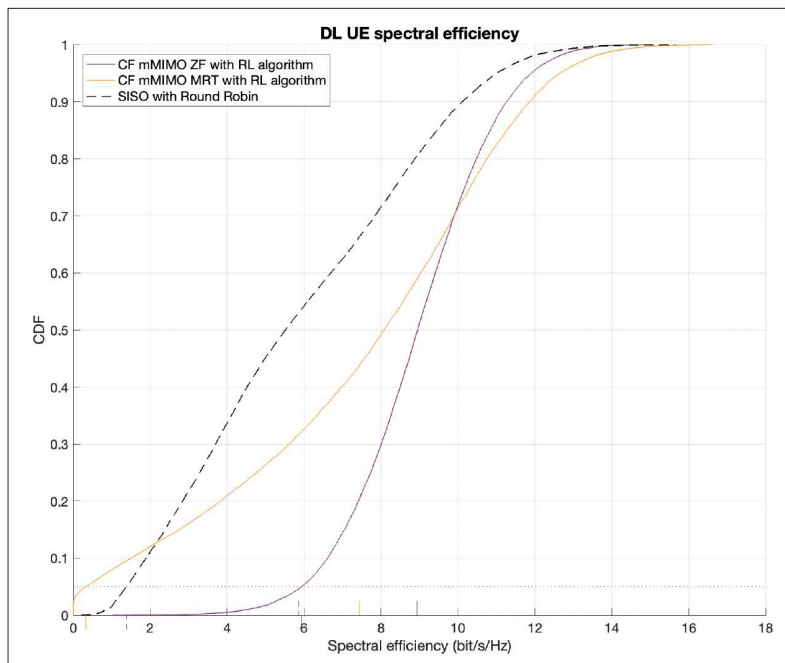


FIGURE 6. DL system spectral efficiency performance of the proposed Q-learning algorithm.

RZF in real system are not obvious. Also the MRT, which is easier in implementation because it does not require precoder coordination between the APs, may be further improved for example with use of the more complex Deep RL algorithms.

CONCLUSION

This article presents a novel CF-based solution that allows the significant scaling up of the RUs in a cost-effective manner, by exploiting the application of the distributed CF concept and of the serial fronthaul approach, while contributing innovative functionalities to the O-RAN Alliance architecture project. In parallel, in the fronthaul/midhaul segments our approach aims to radically increase the flexibility of optical access architectures for Beyond-5G/6G Cell Site connectivity via different levels of fixed-mobile convergence. The presented Q-learning-based evaluation results show that the proposed networking configuration is efficient and can be incorporated into the era of 6G converged-optical-wireless-based networks.

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ADDITIONAL READING

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BIOGRAPHIES

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